

# ThRACE-1

## Mission Statement

Thermodynamic and Radiation Analysis CubeSat Experiment

| Document        | Revision | Status | Date      |
|-----------------|----------|--------|-----------|
| ThRACE-1-MS-001 | Rev 1.2  | FINAL  | July 2026 |

## 1. MISSION OVERVIEW

ThRACE-1 is a student-designed and built 2U CubeSat mission with the objective of determining how good commercial grade electronics can perform in space, as well as characterizing the thermal and radiation environment in Earth orbit compared to known measurements and pre-flight simulations. The satellite is designed for a 500km, 45 degree inclination orbit. It measures particle radiation flux, surface temperatures across all faces, and measures the magnetic field in earth orbit. All data is georeferenced. The commercial electronics on board are an off-the-shelf MEMS IMU and GNSS receiver. They will be tested to see how they work at high radiation and velocity.

The mission is designed and constructed almost entirely from scratch. The on-board computer is a space-modified derivative of the student's Mixer V2 high-power-rocketry flight computer, built around the STM32H753 microcontroller, alongside a custom electrical power system and a DIY ground station with a self-built antenna tracking system. The OBC is mounted on a central sled in the middle of the cubesat, and aggregates the thermal and payload sensors directly rather than through a separate sensor board. Commercial off-the-shelf hardware is used only where custom design would not meaningfully improve the mission outcome. ThRACE-1 serves as both a scientific instrument and a demonstration of end-to-end spacecraft engineering capability.

## 2. MISSION OBJECTIVES

- **OBJ-1** Measure radiation dose rate as a function of orbital position using a Geiger-Müller tube, producing a georeferenced radiation map along the ground track.
- **OBJ-2** Characterize the thermal environment of a 2U CubeSat across sunlight and eclipse cycles by logging surface temperatures from PT1000 RTD sensors on each face.
- **OBJ-3** Map the local geomagnetic field along the orbital path using a three-axis magnetometer and compare measured values to the International Geomagnetic Reference Field (IGRF) model.
- **OBJ-4** Validate the satellite's passive radiation shielding design (2mm Al6061 + 1mm polyethylene liner) by comparing measured dose rates to pre-flight model predictions.
- **OBJ-5** Validate the satellite's thermal control design (internal MLI blankets, bottom-face radiator) by comparing measured temperature cycles to Fusion 360 thermal simulation predictions.
- **OBJ-6** Demonstrate reliable operation of a custom Mixer V2-derived STM32H753 OBC and custom EPS in the space environment for the duration of the mission.
- **OBJ-7** Monitor radiation-induced drift in the CubeADCS reference IMU over the mission lifetime, correlating drift events with measured particle flux from the Geiger-Müller tube.

- **OBJ-8** Characterize the space radiation susceptibility of a commercial-grade MEMS IMU (LSM6DSO) by comparing its output against the reference CubeADCS IMU across varying radiation environments.
- **OBJ-9** Evaluate the on-orbit performance of a consumer-grade GNSS receiver (u-blox SAM-M10Q) against a space-rated receiver, characterizing fix availability and accuracy at orbital velocity.

3. ORBIT AND MISSION PARAMETERS

| Parameter                  | Value                                                               |
|----------------------------|---------------------------------------------------------------------|
| Form Factor                | 2U CubeSat (10 × 10 × 20 cm)                                        |
| Target Orbit               | Low Earth Orbit (LEO)                                               |
| Target Altitude            | ~500 km                                                             |
| Inclination                | 45°                                                                 |
| Orbital Period             | ~95 minutes                                                         |
| Eclipse Fraction           | ~37% per orbit (~35 min eclipse / ~60 min sunlight)                 |
| Estimated Orbital Lifetime | ~7–12 years to natural reentry (compliant with 25-year rule)        |
| Target Mission Duration    | 12 months minimum, 24 months goal                                   |
| Estimated Mass             | ~1.3 kg estimate (TBD pending detailed mass budget; 2.66 kg limit)  |
| Avg Power Generation       | ~2.5 – 4W from 1 fixed GaAs face + 2 deployable wings, sun-pointing |
| Launch Vehicle             | TBD — NASA CSLI (primary) or SpaceX Transporter rideshare (backup)  |

4. SPACECRAFT OVERVIEW

ThRACE-1 is built around a 2mm Al6061 aluminum chassis with a 1mm high-density polyethylene inner liner providing passive radiation shielding. Internal MLI blankets provide thermal insulation, and the bottom face of the satellite is left uninsulated and coated with a high-emissivity white radiator finish (e.g. AZ-93) to serve as a dedicated heat-rejection surface. A single Geiger-Müller tube and its 400V high-voltage supply are isolated in a dedicated payload volume at the radiator end of the satellite, physically separated from the avionics and radio to prevent electromagnetic interference. Subsystems are arranged along the long axis: radio and OBC at the antenna end, the CubeADCS unit (approximately half a U in the sun-pointing configuration) centrally, the battery and EPS between, and the isolated Geiger volume at the radiator end. The satellite is designed to be compliant with the CubeSat Design Specification (CDS) Rev 14.

| Subsystem | Approach | Implementation                                                                                                                                                                                             |
|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBC       | Custom   | Mixer V2-derived STM32H753 (Cortex-M7), reduced clock for thermal margin, dual GD25Q32E datalog flash, RTC with supercap backup, USB-C programming/data, hardware crypto for uplink auth, custom KiCad PCB |

|                       |                      |                                                                                                                                              |
|-----------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EPS</b>            | <b>Custom</b>        | MPPT solar input, battery management, two-stage regulation (12V→5V→3.3V), per-rail current sensing and latch-up protection, custom KiCad PCB |
| <b>ADCS</b>           | <b>COTS</b>          | CubeSpace CubeADCS — reaction wheels + magnetorquers, sun-pointing mode; reference IMU for payload experiment                                |
| <b>COMMS</b>          | <b>COTS</b>          | Endurosat UHF Type II transceiver, 9.6 kbps AX.25 / CSP, deployable turnstile antenna, JST-GH interface                                      |
| <b>Structure</b>      | <b>COTS + Custom</b> | Commercial 2U Al6061 chassis + custom PE liner, MLI, deployable solar wings, isolated Geiger payload volume                                  |
| <b>Flight SW</b>      | <b>Custom</b>        | FreeRTOS task architecture evolved from Mixer V2 firmware, autonomous FDIR, watchdog with reboot recovery, store-and-forward telemetry       |
| <b>Ground Station</b> | <b>Custom</b>        | DIY SatNOGS rotator, Arrow Antenna II, LNA, RTL-SDR, Direwolf, InfluxDB, Grafana, Flask + Cesium 3D visualization                            |

## 5. SCIENCE & TECHNOLOGY PAYLOAD

| Instrument                                                          | Measurement                                       | Science / Technology Value                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Geiger-Müller Tube (isolated 1U payload volume)</b>              | Particle radiation flux (CPM, $\mu\text{Sv/hr}$ ) | Maps radiation dose rate vs. orbital position. Identifies South Atlantic Anomaly crossings. Validates passive shielding predictions. Provides radiation reference for IMU experiments.                                                                                                                          |
| <b>PT1000 RTD × 6 (one per face)</b>                                | Surface temp. ( $^{\circ}\text{C}$ , all faces)   | Characterizes thermal environment across full orbital cycle. Validates MLI blanket and radiator face thermal model predictions.                                                                                                                                                                                 |
| <b>3-Axis Magnetometer (via CubeADCS)</b>                           | Magnetic field vector (nT)                        | Maps local geomagnetic field along ground track. Compares to IGRF model. Secondary use: magnetorquer desaturation reference.                                                                                                                                                                                    |
| <b>Dual-IMU Experiment (CubeADCS reference + LSM6DSO test unit)</b> | Gyro/accel bias drift, temp sensitivity           | Characterizes radiation susceptibility of an untested consumer MEMS IMU against a flight-proven reference, correlated with Geiger dose data. Novel dataset — minimal public radiation characterization exists for consumer MEMS parts.                                                                          |
| <b>Dual-GNSS Experiment (SkyTraq Venus838 + u-blox SAM-M10Q)</b>    | Position, velocity, fix availability              | Space-rated SkyTraq (UART, external patch antenna) provides operational position data, superior to TLE propagation for science georeferencing. SAM-M10Q (I2C, integrated antenna) flown as test subject to characterize consumer GNSS behavior at orbital velocity, including the COCOM 500 m/s velocity limit. |

|                         |                   |                                                                                                                  |
|-------------------------|-------------------|------------------------------------------------------------------------------------------------------------------|
| Solar Irradiance Sensor | Solar flux (W/m²) | Validates solar panel power generation model. Correlates eclipse transitions with thermal and power system data. |
|-------------------------|-------------------|------------------------------------------------------------------------------------------------------------------|

6. MISSION SUCCESS CRITERIA

Minimum Success

- Satellite survives launch and deploys from P-POD successfully
- UHF beacon received by ground station or SatNOGS network within first 3 orbits
- At least one full telemetry downlink session completed with all sensor readings present
- Satellite operates for a minimum of 30 days without unrecoverable failure

Full Mission Success

- All nine mission objectives (OBJ-1 through OBJ-9) met with continuous data collection for 12+ months
- Radiation map covering 100+ orbits with South Atlantic Anomaly clearly identifiable in dataset
- Measured face temperatures match thermal simulation predictions within ±10°C RMS
- Measured radiation dose within ±25% of shielding model prediction
- Dual-IMU dataset sufficient to publish drift-vs-dose correlation for the LSM6DSO
- SAM-M10Q fix availability characterized across at least 50 orbits
- Dataset published to Zenodo with DOI and shared with the SatNOGS and amateur satellite communities
- Real-time mission data publicly accessible via Grafana dashboard and Cesium 3D visualization

7. HIGH-LEVEL MISSION TIMELINE

| Phase   | Description                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------------------|
| 2026 H1 | Ground station build. Mixer V2 design and flight (STM32H743 learning platform). ADCS coursework.                   |
| 2026 H2 | ThRACE-1 OBC derived from Mixer V2. Custom EPS design and fabrication. Payload board design. Flight software.      |
| 2027 H1 | Full system FlatSat test (including 48-hour soak). Chassis integration. Environmental testing. Prototype complete. |
| 2027 H2 | Portfolio documentation. College applications. FCC experimental license groundwork. CSLI proposal draft.           |
| 2028+   | University enrollment. Faculty advisor. CSLI submission. Flight unit upgrades. TVAC/vibration. Launch campaign.    |

8. DOCUMENT INFORMATION

|             |                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Author      | Cortland Familian                                                                                                                                                    |
| Mission     | ThRACE-1 — Thermodynamic and Radiation Analysis CubeSat Experiment                                                                                                   |
| Document ID | ThRACE-1-MS-001                                                                                                                                                      |
| Revision    | Rev 1.2 — OBC confirmed as Mixer V2-derived STM32H753; photodiode sun-sensor array dropped; single Geiger tube; sensor board consolidated onto OBC; dual GNSS locked |
| Repository  | <a href="https://github.com/mixertechdev/ThRACE">https://github.com/mixertechdev/ThRACE</a>                                                                          |

9. REVISION HISTORY

| Revision | Date       | Changes                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rev 1.0  | April 2026 | Initial draft as TRACE-1                                                                                                                                                                                                                                                                                                                                                                                         |
| Rev 1.1  | May 2026   | Renamed to ThRACE-1. OBC changed to Mixer V2-derived STM32H743. Added OBJ-7/OBJ-8 (dual-IMU radiation experiment), OBJ-9 (dual-GNSS experiment). Geiger tube isolated in dedicated payload volume. Added orbital lifetime estimate and revision history.                                                                                                                                                         |
| Rev 1.2  | July 2026  | OBC confirmed as STM32H753 (crypto for uplink auth), mounted central above ADCS. Supplementary photodiode sun-sensor array dropped (CubeADCS provides sun sensing). Payload sense board eliminated; thermal/payload sensors aggregate directly onto OBC. Single Geiger tube confirmed. Dual GNSS locked (SkyTraq UART + SAM-M10Q I2C, separate antennas). Radiator coating and long-axis stack layout specified. |

This document is a living reference and will be updated as design decisions are finalized. Refer to the version number in the header to confirm currency.